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Article

Self-Calibrating Gas Pressure Sensor with a 10-Decade Measurement Range

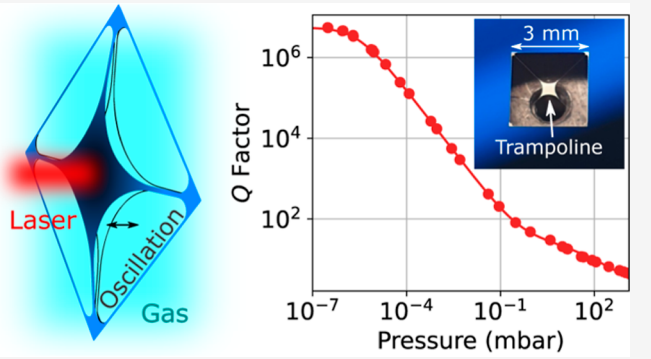
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ABSTRACT: Recent years have seen a rapid reduction in the intrinsic loss of nanomechanical resonators (i.e., chip-scale mechanical oscillators). As a result, these devices become increasingly sensitive to the friction exerted by the smallest amounts of gas. Here, we present the pressure-dependency of a nanomechanical trampoline resonator's quality factor Q over 10-decades, from 10^{-7} to 10^{3} mbar. We find that the measured behavior is well-described by a model combining analytical and numerical components for molecular and viscous flow, respectively. This model relies exclusively on design and typical material parameters, together with measured values of intrinsic resonance frequency f_{in} and quality factor Q_{in}. Measuring f_{in} and Q_{in} at a pressure $<10^{-7}$ mbar self-calibrate our sensor over its entire measurement range. For a trampoline's fundamental out-of-plane vibrational mode, the resulting deviation between measured and simulated pressure dependencies of the quality factor and resonance frequency is within 15 and 4%, respectively. The resulting error for pressure values inferred from quality factor and frequency measurements is <10% for pressures between $\sim 10^{-6}$ and $\sim 10^{-1}$ mbar, and <25% for the complete 10-decade measurement range. Exceptions are two outliers with increased measurement errors, which might be related to the limited accuracy of our commercial pressure gauge. Based on investigations with helium, we demonstrate the potential for extending this sensing capability to other gases, thereby highlighting the practical use of our sensor.



KEYWORDS: *optomechanics, nanomechanics, pressure sensing, vacuum technology, nanofluidics*

INTRODUCTION

Establishing and accurately measuring low gas pressures is a prerequisite for cutting-edge technologies such as semiconductor fabrication, quantum computing, and fundamental science. Typical operating pressures P span ultrahigh vacuum ($<10^{-8}$ mbar) to ambient pressure ($\sim 10^{3}$ mbar), which currently requires two to three different sensors for a complete coverage.[1,2] Depending on their measurement range, these devices rely on different physical phenomena, such as the transport of thermal energy by, or the ionization of, residual gas particles, or the pressure-induced deflection of a diaphragm.[1] While the latter approach represents a direct pressure measurement, the former two enable only an indirect determination, which involves gas-species-dependent properties such as the atomic mass or the specific heat. Consequently, the implementation of transition regions between individual sensors and a separate calibration for each type of gas is required. This adds complexity and cost and can limit the achievable measurement accuracy.

Recent advances in nanofabrication have enabled the realization of compact absolute[2] and differential[3,4] pressure sensors with wide measurement ranges, hinting at the possibility of covering ultrahigh vacuum to ambient pressure with a single sensor. However, realizing such a device still requires increasing measurement range and accuracy, compared to demonstrated sensors, and establishing a physical model or a calibration function to describe the sensor's pressure dependency. To this end, nano- and micromechanical resonators are promising candidates.[5] A whole variety of these devices have been demonstrated with a sensitivity partially or fully covering the range of 10^{-3} to 10^{3} mbar.[6−21] Furthermore, it has been pointed out that increasing the intrinsic quality factors Q_{in}, e.g., by optimizing the device geometry, can significantly extend their measurement range toward lower pressure ranges.[22] Recently, the sensitivity of a phononic

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crystal membrane resonator has been demonstrated in the range from 10^{-7} to 2×10^{-5} mbar, enabled by its ultrahigh $Q_{in} \sim 10^8$.[23]

The measurement principle of resonant mechanical sensors relies on a pressure-dependency of either their frequency f_m, quality factor Q, or amplitude.[24−26] In the free-molecular flow (FMF) regime, where the gas particles do not interact with each other, Q follows[24]

$$Q = (Q_{in}^{-1} + Q_{FMF}^{-1})^{-1} \quad (1)$$

Here, Q_{in} is the intrinsic quality factor of the resonator and Q_{FMF} describes the pressure dependency resulting from collisions with gas particles. The latter is given by

$$Q_{FMF} = \frac{f_{in}\rho h}{P}\sqrt{\frac{\pi^3 k_B T_{gas}}{8 m_{gas}}} \quad (2)$$

with intrinsic resonance frequency f_{in}, mass density ρ, and thickness h of the resonator, as well as temperature T_{gas} and atomic mass m_{gas} of the gas. When the pressure is continuously increased, the interaction between gas particles becomes important, such that the gas acts as a fluid. This has two main consequences: first, viscous damping is the dominant loss mechanism acting on the resonator. Second, a fraction of the gas co-oscillates with the resonator, which contributes an additional load m_{add} to its intrinsic mass m_{eff}, thereby reducing its oscillation frequency according to[27,28]

$$f_m = f_{in}[1 + m_{add}/m_{eff}]^{-1/2} \quad (3)$$

The specific pressure at which the transition between FMF and viscous flow (VF) regimes occurs depends on the properties of both the device and the fluid.[29,30] Simplified models, such as the two-dimensional (2D) oscillating cylinder model,[27] are used to describe the viscous damping of cantilevers and beams[a].[31,32] Resonators with geometries corresponding to arrangements of multiple beams have been modeled by representing the device as a string of spheres.[6,33] This approach makes use of the well-known analytical solution for a sphere oscillating in a viscous fluid.[34] An approximate analytical model describing gas-induced loss over the entire range, from FMF to VF, and the intermediate transitional flow regime, has been established and effectively applied to beams and cantilevers.[35,36] To overcome the limitations of analytical models, a recent article[37] describes a finite element method (FEM) for modeling f_m and Q of cantilevers and beams in the VF regime.

Here, we investigate gas pressure sensing with ultrahigh-Q mechanical trampoline resonators,[38,39] featuring $Q_{in} \sim 10^7$. For the fundamental out-of-plane mode of the best device, we observe a continuous change in Q from $Q = 5 \times 10^6$ at 10^{-7} mbar to $Q = 7$ at 10^3 mbar. This corresponds to a sensitivity range of 10-decades, which, to the best of our knowledge, is unprecedented. Depending on the trampoline's geometry, vibrational mode, and investigation of either air or helium, a transition between FMF and VF occurs between 1 and 100 mbar. We find that the model

$$Q_{mod} = [Q_{in}^{-1} + (Q_{FMF} + Q_{VF})^{-1}]^{-1} \quad (4)$$

well-describes the observed behavior over the investigated range, where Q_{VF} represents the P-dependent Q in the VF regime. In the first step, we express Q_{VF} as a two-parameter fit function, which provides limited agreement with the data. In a second step, to establish a more accurate model of viscous damping, we simulate the interaction of our device with the surrounding fluid via FEM in Comsol Multiphysics, similar to ref 37. By substituting Q_{VF} in eq 4 with the simulated values, we find an overall agreement between the data and the combined analytical-FEM model within $\leq\pm$ 15%. The only model inputs are design and typical material parameters, together with f_{in} and Q_{in}. Self-calibration of our sensor, from near ultrahigh vacuum to ambient pressure, is implemented by measuring f_{in} and Q_{in} at a pressure $<10^{-7}$ mbar. This alleviates the need for a resource-demanding calibration by comparison with a reference gauge.[40] In addition to Q, we investigate the P-dependency of the trampolines' resonance frequencies f_m in the VF regime. We find a significant reduction with respect to f_{in} of up to 38% at ambient pressure. The resonance frequency simulated with our FEM model matches the data within 4% over 3-decades, between 1 and 1000 mbar. The error associated with pressure values inferred from Q and f_m measurements is determined to be <10%, for pressures between $\sim 10^{-6}$ and $\sim 10^{-1}$ mbar, and <25% between $\sim 10^{-7}$ and $\sim 10^3$ mbar. For two pressure values lying between 1 and 10 mbar, we find an increased error of up to 57%. We argue that the deviations between data and models for both Q and f_m, particularly the two outliers, might be dominated by a commercial pressure gauge. For one of our devices, we also use helium gas in addition to air, finding equally good agreement between data and fit functions. Finally, we discuss possible routes for extending the measurement range toward ultrahigh vacuum, for which currently only ionization gauges[41] and cold atoms[42] provide adequate sensitivities.

TRAMPOLINE PRESSURE SENSOR

Measurement Setup. Figure 1A shows a schematic of the experimental setup used for investigating the trampoline pressure sensor. Here, the device is placed inside an ultrahigh vacuum chamber, in which the pressure is set between 10^{-8} and 10^3 mbar. This is realized by injecting controlled amounts of gas (e.g., helium) via a gas-regulating valve (Pfeiffer EVR 116) and a corresponding controller (Pfeiffer RVC 300). Furthermore, the opening of a gate valve, located in front of a turbo pump, is adjusted to set the effective pumping speed. The pressure inside the chamber is measured with a Bayard-Alpert/Pirani combination gauge (Pfeiffer PBR 260), between 10^{-8} and 10 mbar, and a capacitive gauge (Pfeiffer CMR 361), between 10 and 10^3 mbar. To measure the oscillation amplitude for some of the trampoline's out-of-plane vibrational modes (the fundamental mode is shown), a Michelson interferometer is used. Here, an incident laser, having a wavelength of 1064 nm, is split into two beams of equal intensity. These beams are reflected by either a mirror or the trampoline and subsequently interfere at a photodiode (PD). The PD's output signal, which is modulated at f_m, is measured either with a lock-in amplifier or a spectrum analyzer.

Figure 1B shows a drawing of the trampoline design,[38,39] where the device with lateral extent w comprises a central pad of width d, which is suspended by four tethers of width a. Corner fillets, defined via circular segments of radius r, connect the trampoline to the supporting silicon chip. For the work presented in the following, we investigated two trampolines, which were obtained from Norcada Inc.[43] They are made out of a silicon nitride thin film, having a high tensile stress of $\sigma = 1$ GPa and $\rho = 3000$ kg/m^3, which is freely suspended from a silicon chip. The first device, referred to as trampoline i, is

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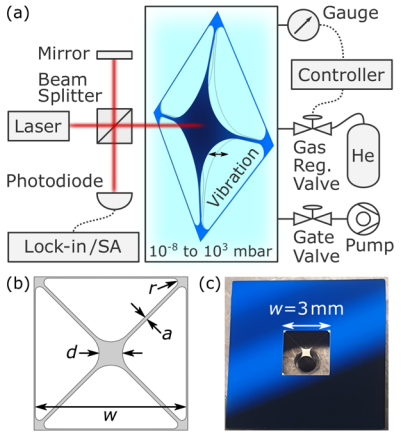


Figure 1. Pressure sensing with a nanomechanical trampoline resonator. (A) Trampoline is situated inside an ultrahigh-vacuum chamber. Thermal energy in the environment or resonant excitation via a piezo actuator (not shown) cause the trampoline to vibrate at its characteristic resonance frequencies f_m (measured fundamental mode shown). Surrounding gas impacts the vibration via a frictional force and, in the viscous flow regime, an increased effective mass. The trampoline's vibration is measured with a Michelson interferometer comprising a laser, a beam splitter, a mirror, and a photodiode connected to a lock-in amplifier or a spectrum analyzer (SA). For our proof of principle, the pressure in the chamber is monitored with commercial gauges and adjusted through a gas regulating valve, both connected to a controller. In combination with adjusting the effective pumping speed of a turbo pump, via a gate valve, the pressure in the chamber is set between 10^{-8} and 10^{3} mbar. (B,C) Trampoline design and investigated trampoline resonator with window size w = 3 mm, pad diameter d = 500 μm, tether width a = 2 μm, and corner fillet radius r = 75 μm.

characterized by w = 1 mm, d = 150 μm, a = 21.5 μm, r = 25 μm, and h = 100 nm. The dimensions of the second device, referred to as trampoline ii, are w = 3 mm, d = 500 μm, a = 2 μm, r = 75 μm, and h = 80 nm. Trampoline ii is shown in Figure 1C, where the silicon nitride film appears blue on top of silicon and white in regions where it is freely suspended. Note that the given mass density and the specified thicknesses are used for all models, as presented in Figures 3−5 (i.e., all oscillation modes of the same device use exactly the same parameters). Possible variations in the membrane's thickness of 5% and in its mass density of 10% (see, e.g., refs 23 and 44) might contribute to the systematic errors, as presented in Figure 5.

Sensor Readout. To measure a trampoline's f_m and Q, we perform mechanical ringdowns of its oscillation amplitude for $P \lesssim 10^{-2}$ mbar. At higher pressures, due to the ringdown's short duration ($\lesssim$0.1 s for the oscillation amplitude to decay by a factor $1/e$), we assess the noise spectra related to thermally driven oscillations. A mechanical ringdown is obtained by resonantly exciting an oscillation mode via a piezo actuator, which is connected to the lock-in amplifier's output (we have used piezos placed either inside or outside the UHV chamber; effective excitation with the external piezo, which is mounted to the same optical table as the trampoline, is facilitated by the trampolines' exquisite force sensitivity[38]). Upon switching off the piezo drive, the trampoline's oscillation amplitude follows an exponential decay until it reaches the thermal noise floor, where it undergoes random (i.e., Brownian) motion, as shown in Figure 2A. Here, data sets for trampoline i's fundamental oscillation mode at air pressures of 2×10^{-7}, 10^{-4}, and 10^{-3} mbar are displayed, corresponding to quality factors of 9×10^6, 2×10^6, and 3×10^5, respectively. The Q values are obtained from fitting $[(a_1\, e^{-\pi f_m t/Q})^2 + b_1^2]^{-1/2}$ (lines in Figure 2A), with fit parameters a_1, a_2, time t, and intrinsic resonance frequency f_{in} = 173.6 kHz, to the data.[38] Here, $\tau = Q/(\pi f_{in})$ corresponds to the ringdown time.

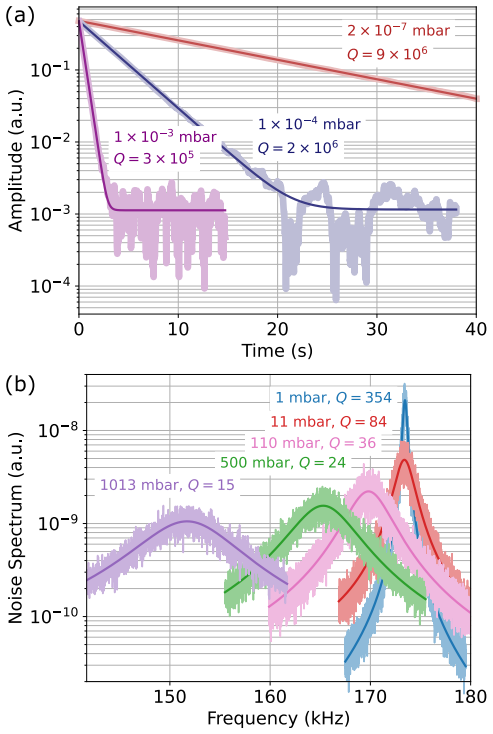


Figure 2. Measuring mechanical quality factor Q and resonance frequency f_m at different air pressures P (shown plots represent the fundamental mode of trampoline i, as described in the main text). For P < 0.1 mbar, the gas mainly impacts the resonator's Q. Increasing P above this level causes a reduction in f_m, due to gas particles vibrating together with the resonator, thereby increasing its effective mass. (A) Typical ringdown of oscillation amplitude vs time t, occurring after switching of piezoexcitation, at three different pressure values. Q is extracted from the fit function $[(a_1\, e^{-\pi f_m t/Q})^2 + b_1^2]^{-1/2}$ (solid line), with initial amplitude a_1 and thermal noise level b_1 (Q, a_1, and b_1 are fit parameters). (B) Typical thermal displacement noise spectra for five different pressure values. For $P \gtrsim 10^{-2}$ mbar, Q and f_m are extracted from fitting a Lorentzian function (solid line) $a_2\{[4\pi^2(f_m^2 - f^2)]^2 + (4\pi^2 f_m f/Q)^2\}^{-2} + b_2$, with fit parameters Q, f_m, a_2, and b_2.

Figure 2B shows the displacement noise spectra of trampoline i's fundamental out-of-plane mode for five different air pressures. Q and f_m are obtained from fitting a Lorentzian (lines in Figure 2B) $a_2\{[4\pi^2(f_m^2 - f^2)]^2 + (4\pi^2 f_m f/Q)^2\}^{-2} + b_2$, with fit parameters a_2 and b_2, to the data. Up to ∼10 mbar, the effect of increasing P is limited to broadening the resonance, thereby decreasing Q. Further raising P additionally causes a reduction in the resonance frequency, thereby indicating the FMF−VF transition, which is consistent with eq 3. At ambient pressure (1013 mbar), $f_{in} - f_m$ = 21 kHz, which corresponds to 13%.

The time it takes to obtain a pressure reading with our sensor is $\sim\tau$, for ringdown measurements, and $\sim 100\tau$ for spectral measurements. In the longest case, corresponding to trampoline ii at base pressure $\sim 10^{-7}$ mbar, the measurement time is ∼100 s.

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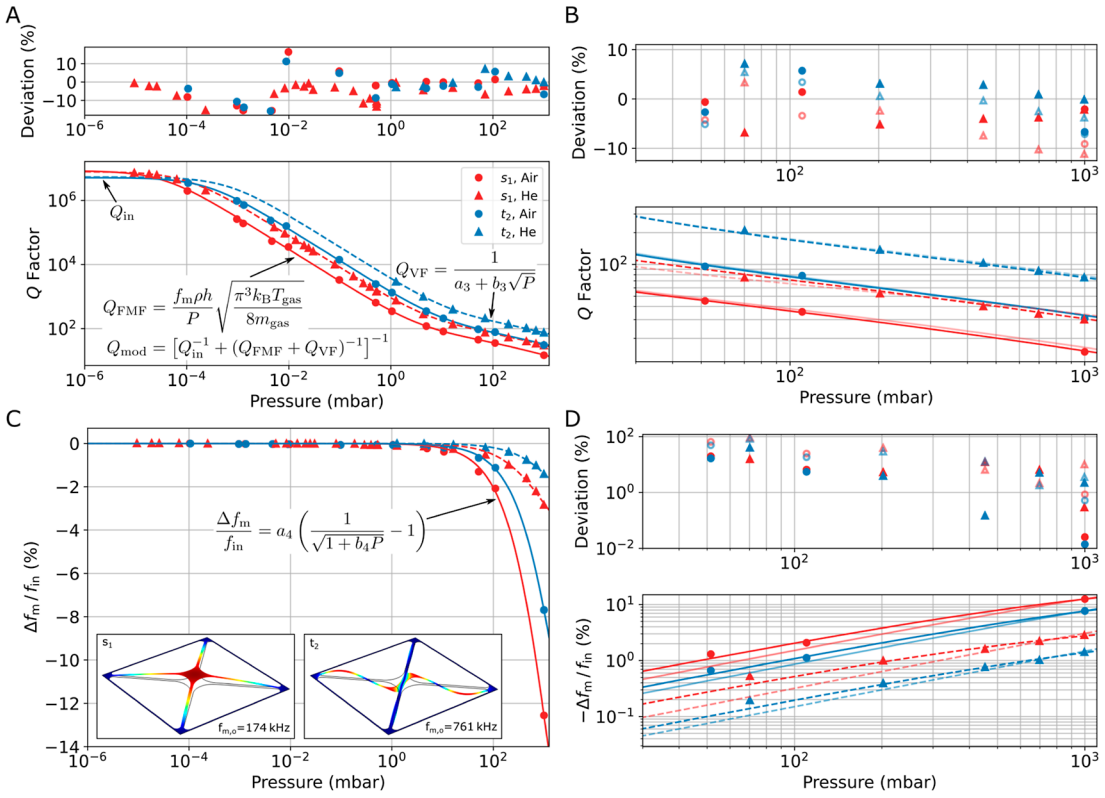


Figure 3. Dependency of mechanical quality factor Q and resonance frequency f_m on air and helium pressure for trampoline i. (A, lower) Q as a function of pressure for s_1 (red) and t_2 (blue) modes (see insets of panel (c) for corresponding mode profiles (simulated with Comsol) and intrinsic resonance frequencies f_{in} (measured)). Measured values Q_i (average of four-seven individual measurements) are represented by circles and triangles for air and helium, respectively. Solid and dashed curves show the corresponding model functions $Q_{mod}(P)$, with intrinsic quality factor Q_{in}, free-molecular flow model Q_{FMF}, and viscous flow fit function Q_{VF}. Explicit expressions are given in the plot, with mass density ρ = 3000 kg/m^3 and thickness h = 100 nm of trampoline i, temperature T_{gas} = 293 K and masses m_{gas} = {4.81, 0.66} × 10^{-27} kg of {air, He} particles, and fit parameters a_3 and b_3. Transition between free-molecular and viscous flow regimes occurs between ∼1 and ∼100 mbar, depending on mode and gas species. (A, upper) Relative deviation of measured values and model function $Q_i/Q_{mod}(P_i) - 1$. (B) Detail of (A), showing only the viscous flow regime. Light red and blue curves (lower) represent an analytical model for viscous damping of an oscillating string of spheres, where the sphere radius is a fit parameter (see the main text for details). Corresponding deviation with respect to the measured values is represented by light open markers (upper). (C) Relative change in frequency, $\Delta f_m/f_{in} = f_m/f_{in} - 1$, as a function of P. Solid and dashed curves represent fit functions (expression given in the plot) with fit parameters a_4 and b_4. (D, lower) Detail of (C), showing only the viscous flow regime. Light red and blue curves (lower) represent an analytical model for viscous damping of a cantilever, with a single size- and mode-shape-dependent fit parameter (see the main text for details). (D, upper) Solid (open) markers represent deviations between measured values and the two-parameter fit function, given in (C), (between measured values and the cantilever model).

Pressure Dependency of Q and f_m. Figure 3A (lower) shows Q as a function of air or helium pressure for the first symmetric (s_1) and the second torsional (t_2) mode of trampoline i. Related mode profiles and intrinsic resonance frequencies simulated in Comsol are shown in the insets of Figure 3C. Measured values for s_1 (t_2) are represented by red (blue) dots and triangles for air and helium, respectively. Each Q value is obtained by averaging four to seven individual measurements. The corresponding uncertainties are typically a few percent, where fitting errors appear to be a main limitation. This has been verified by fitting different sections of ringdown curves and noise spectra. In the best scenario, a standard deviation of 0.5% has been obtained, for a set of six subsequent ringdowns taken at base pressure P_{base} = 5.4 × 10^{-8} mbar. Here, for each ringdown, the trampoline was driven to exactly the same oscillation amplitude, exactly when the previous decay reached the onset of the thermal noise floor. This indicates the potential for high-precision pressure measurements with nanomechanical oscillators. For other ringdown measurements, showing larger deviations, the maximum oscillation amplitude was not well controlled, and part of the thermal noise floor was included in the fit. The model function Q_{mod}, corresponding to each data set, is given by eq 4. They are shown as solid and dashed curves for air and helium, respectively. Here, Q_{in} corresponds to the quality factor measured at P_{base}, which is 8.0 × 10^6 and 5.1 × 10^6 for s_1 and t_2, respectively. These values are about 30% lower than the ones simulated in Comsol, for a limitation due to bending loss,[45,46] under the assumption of a typical material quality factor of 5000 for a 100 nm thick SiN membrane.[47] This discrepancy is consistent with studies of similar devices,[48] which point out additional loss due to radiation of elastic waves to the silicon chip as a possible cause. Q_{FMF} is given by eq 2, with T_{gas} = 293 K and m_{gas} = {4.81, 0.66} × 10^{-27} kg for {air, helium} particles. The VF component is represented by a fit function, $Q_{VF} = (a_3 + b_3\sqrt{P})^{-1}$, with fit parameters a_3 and b_3. It represents a generalized version of the expression for a viscously damped cantilever, assumed to consist of a string of spheres (SOS), which is given by[12,33]

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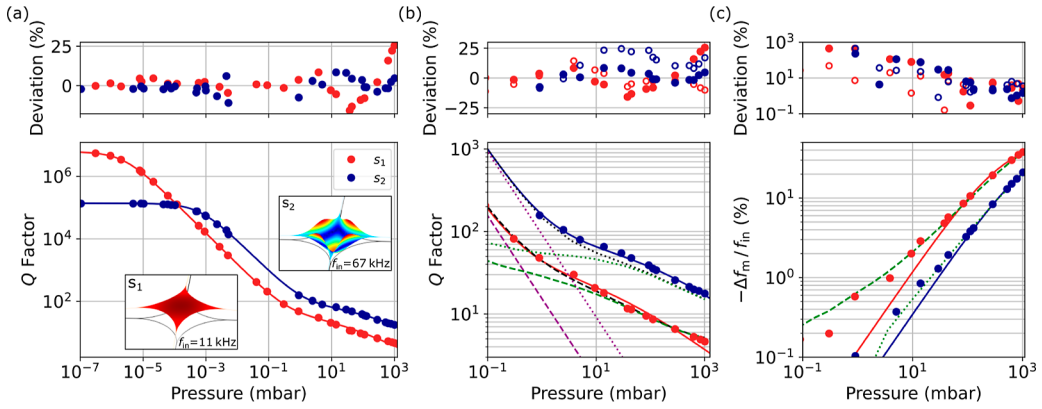


Figure 4. Self-calibrating pressure sensing over 10-decades of air pressure with trampoline ii. (A) Similar to Figure 3A: the bottom panel shows Q VS P for s_1 (red) and s_2 (dark blue) modes, with simulated mode profiles (Comsol) and intrinsic resonance frequencies (measured) shown in the insets. Solid lines represent semiempirical model functions (eq 4; parameters given in the main text), which include two fit parameters. Deviations between measured values and model functions are shown in the upper panel. (B) Detail of (A) showing the top four pressure decades. Purple, green, and black curves show the analytical model for free-molecular flow, Q_{FMF}, the Comsol model for viscous flow, Q_{COM}, and the combined model, $\tilde{Q}_{mod}(P) = [Q_{in}^{-1} + (Q_{FMF} + Q_{COM})^{-1}]^{-1}$, which rely exclusively on known membrane and gas parameters. Here, dashed (dotted) curves are for s_1 (s_2). Open markers in the upper panel show deviations between measured values and $\tilde{Q}_{mod}$, which are within 15% for s_1 over the investigated range. (C) Similar to Figure 3D: relative change in f_m as a function of P. Solid curves represent fit functions (expression given in Figure 3C). Green curves are obtained with Comsol.

$Q_{SOS} = 4\rho h D f_{in}/[6\mu + 3D\sqrt{\pi\mu(m_{gas}/k_B T)f_{in}P}]$, with dynamic viscosity of the gas μ and sphere diameter D.

The onset of the pressure dependency occurs from 10^{-5} to 10^{-3} mbar and the transition between molecular and VF regimes in the range from 1 to 100 mbar, both depending on resonance mode and gas species. Our model corroborates this dependency, as both transitions are determined by the membrane properties ρ, h, and f_{in}, together with gas parameters m_{gas} and T. Furthermore, increasing Q_{in} and D moves the onset of Q's pressure dependency and the FMF−VF transition to lower pressures, respectively. While molecular damping (eq 2) depends only on the thickness of the resonator, viscous damping (Q_{SOS}) also involves the lateral extent, represented by D. This hints at a dependency on the geometry of viscous damping affecting a thin-film resonator, as observed in.[29]

Figure 3A (upper) shows the relative deviation $Q_i/Q_{mod}(P_i) - 1$ between data points Q_i, measured at pressures P_i and the corresponding model values. Overall, data and model agree within ±15%, which matches the specified accuracy of our pressure gauge;[49] between 1 mbar and 10^3 mbar, the deviation is within ±10%. The similarity in the deviations of both modes obtained with air in the molecular flow regime hints at the inaccuracy of the commercial pressure gauge as a probable cause for the deviations. In the VF regime, the semiempirical fit function might pose a significant limitation to attainable accuracy. Figure 3B (lower) shows the corresponding pressure range, where, in addition to Q_{mod} a second model function is shown, in light red and light blue for air and helium, respectively. In this function (eq 4), Q_{VF}, is replaced by Q_{SOS}, leaving D as the sole fit parameter. Corresponding values of D_{s_1} = 45 μm and D_{t_2} = 23 μm are obtained for s_1 and t_2, respectively, which lie within trampoline i's tether and pad widths (see above). $D_{s_1} > D_{t_2}$ is consistent with the larger contribution of the pad to the modes' displacement profiles, for s_1 compared to t_2. The deviations between data and Q_{SOS} (light red/blue open symbols) exceed 10%, in the vicinity of ambient pressure for t_2, as shown in Figure 3B (upper), thereby showcasing the advantage of Q_{VF}, which yields deviations <10%.

Figure 3C shows the relative change in the oscillation frequency $\Delta f_m/f_{in}$, with $\Delta f_m = f_m - f_{in}$, over the investigated pressure range. Data sets are shown together with corresponding fit functions $a_4[(1 + b_4 P)^{-1/2} - 1]$, having fit parameters a_4 and b_4. This function is generalized from the expression for a cantilever,[27] for which $a_4 = 1$ and b_4 depend on oscillation mode and gas properties. The plotting style for both modes and gas types is identical to that in Figure 3A. With regard to Figure 3A, it is apparent that a significant decrease in f_m occurs only in the VF regime. The onset of the effect agrees with the FMF−VF transition. Furthermore, it is more pronounced for s_1 compared to t_2, which implies a correspondingly bigger fluid load m_{add}, according to eq 3; in particular, since m_{eff} of s_1 is larger compared to t_2.[38] The frequency decrease is significantly more pronounced for air compared to helium, which is consistent with the corresponding difference in their particle masses.

A detailed analysis of the frequency decrease (Figure 3C), covering the VF regime, is shown in Figure 3D. In addition to the two-parameter fits, this plot also contains fits with $a_4 = 1$ (i.e., of a cantilever type), shown in light red and light blue for air and helium, respectively. Corresponding deviations are shown in Figure 3D (upper), where light colored open symbols represent the deviation between the data and cantilever-type fits. These deviations exceed the ones obtained with two free fit parameters multiple times, for most data points. Between 100 and 1000 mbar, the two-parameter fit matches the data within 10%, while at lower pressures, the deviations are larger.

Self-Calibrated Pressure Sensing. Figure 4A (lower) shows the dependency on air pressure of Q for trampoline ii (see Figure 1C). The first and second symmetric modes, designated as s_1 and s_2, with Q_{in} of 6.1×10^6 and 1.4×10^5, respectively, have been investigated; corresponding mode profiles and intrinsic resonance frequencies, simulated in

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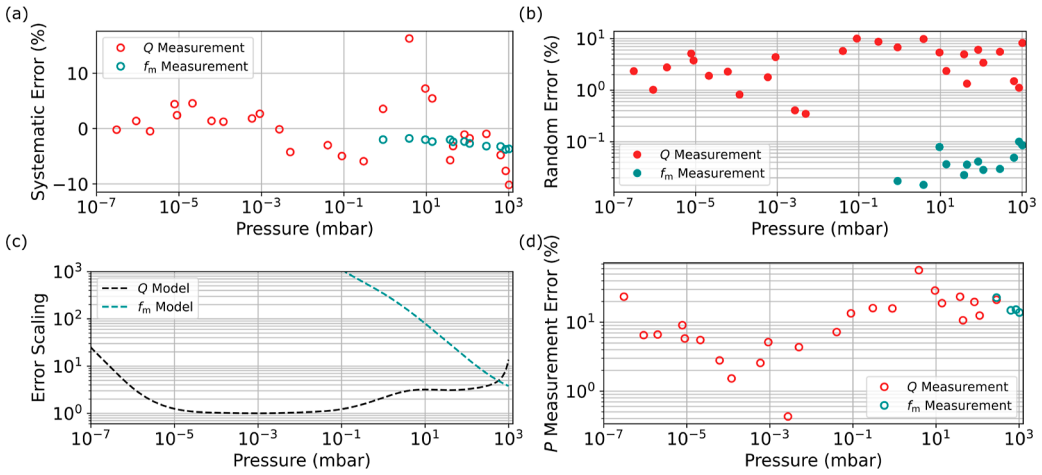


Figure 5. Errors of the air pressure measurement with the s_1 mode of trampoline ii. (a) Systematic errors of measured quality factors Q (red) and resonance frequencies f_m (cyan), corresponding to the relative deviations between data $\{Q_i, f_{m,i}\}$ and respective models $\{\tilde{Q}_{mod}, f_{COM}\}$ (see Figure 4). (b) Random errors of Q and f_m measurements; each data point corresponds to the standard deviation of a set of repeated measurements. (c) Proportionality constants relating the errors in Q and f_m measurements, respectively, to inferred P values (derived based on the model functions shown in Figure 4b,c). (d) Error of the P values inferred from Q and f_m measurements, corresponding to the root sum square of systematic and random errors (shown in panels (a,b), respectively) rescaled by the proportionality constant shown in panel (c). Errors related to the Q and f_m measurements are shown for $P \leq 287$ mbar and $P > 287$ mbar, respectively, where they provide the lowest measurement error.

Comsol, are shown in the insets. Measured values of Q_{in} are lower by a factor 3 compared to the ones simulated in Comsol, for a limitation due to bending loss.[45,46] Here, a typical material quality factor for an 80 nm thick SiN membrane of 4000[47] is assumed. The discrepancy between measured and simulated Q factors is similar to the one reported in ref 50. Here, the authors point out a potential relationship between the increased loss and a static out-of-plane deformation (i.e., *buckling*) of the device. In fact, we see evidence of buckling for trampoline ii when viewing it under a microscope. Optimizing the device geometry might enable avoiding this limitation. Together with the data (dots), associated model functions (lines), given by eq 4, with the two-parameter fit function Q_{VF}, are presented. s_1 shows a pronounced pressure dependency from $\sim 10^{-7}$ to $\sim 10^{3}$ mbar, thereby realizing a 10-decade measurement range, which, to the best of our knowledge, is unprecedented. The extended sensitivity toward lower pressure, compared to trampoline i, is a consequence of trampoline ii's 16-fold lower resonance frequency. Effects on the sensitivity range due to differences in h and Q_{in}, between trampoline i and ii, cancel each other within 14%. The FMF−VF transition occurs between 0.1 and 1 mbar, i.e., at about 10-fold lower pressure compared to s_1 of trampoline i. This appears to be a consequence of trampoline ii's lower resonance frequency and larger spatial extent, compared to trampoline i, which follows from equating Q_{FMF} with the two-parameter fit function Q_{VF} and solving for P. For s_2, a pressure dependency only from $\sim 10^{-4}$ mbar, due to its comparatively low Q_{in}, up to $\sim 10^{3}$ mbar, is evident. The FMF−VF transition of s_2 occurs between 0.4 and 40 mbar. Deviations between the data and model functions are shown in Figure 4A (upper). They are <15% for s_2 over the investigated pressure range. s_1 shows a divergence between data and model for $P > 15$ mbar, which reaches up to 25% at ambient pressure, thereby indicating a systematic deviation from the semiempirical model function Q_{VF}.

To establish a more accurate model of viscous damping acting on the trampoline, we have implemented a corresponding simulation in Comsol by following the approach presented in ref 37 for cantilevers and beams. Here, the trampoline's oscillation, implemented in the structural mechanics module, is coupled to the viscous fluid (governed by Navier−Stokes equations), implemented in the acoustic module. The resulting model includes energy dissipation from the trampoline via both viscous drag and thermal conduction, where the latter is negligible as temperature gradients are insignificant in our case. Figure 4B shows a detail of Figure 4A, focusing on the upper 4-decades of pressure. Here, the simulated quality factors related to viscous damping, Q_{COM}, are shown in green. For s_1 and $P \geq 40$ mbar, the simulation gives excellent agreement with measured values. By replacing Q_{VF} in eq 4 with Q_{COM}, we establish a model, $\tilde{Q}_{mod} = [Q_{in}^{-1} + (Q_{FMF} + Q_{COM})^{-1}]^{-1}$, which covers the investigated pressure range from FMF to VF and relies exclusively on known membrane parameters and gas type, together with f_{in} and Q_{in}. Measuring f_{in} and Q_{in} at $P < 10^{-7}$ mbar self-calibrates the sensor over its entire measurement range. $\tilde{Q}_{mod}$ is represented by the black line, which agrees with the data within 15% over the entire measurement range, indicated by the open red circles in Figure 4B (upper). Consequently, $\tilde{Q}_{mod}$ accurately predicts the pressure range over which the FMF−VF transition occurs. For decreasing pressure, $\tilde{Q}_{mod}$ asymptotically converges to Q_{FMF} (purple). For s_2, Q_{COM} provides systematically larger values (up to 25%), compared to the data. This is considered to follow from marginal geometric imperfections, such as slight out-of-plane buckling of the central pad's rim (indicated in investigations with an optical microscope; could be avoided by a slight design optimization), which are not accounted for in the simulation.

Figure 4C shows $-\Delta f_m/f_{in}$ for trampoline ii in the VF regime. Measured values (dots) are shown together with two-parameter fit functions (red and blue lines; see previous section for details) and $f_{COM}/f_{COM,in} - 1$ (green dashed/dotted lines). Here, f_{COM} and $f_{COM,in}$ are the values simulated in Comsol for the P-dependent resonance frequency in the VF regime and the intrinsic resonance frequency, respectively. For

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s_1 and s_2, the oscillation frequency is reduced by 38 and 21%, respectively, at ambient pressure. From 10^2 to 10^3 mbar, the fit functions match the data within 10%. However, for $P < 10$ mbar, they significantly diverge from measured values, thereby ceasing to provide an accurate description. The Comsol model matches the data over a significantly broader range, providing a deviation within 15% for s_1, from 1 mbar to 10^3 mbar, and within 25% for s_2, from 10 mbar to 10^3 mbar.

Measurement Errors. In the following, we assess the error of the pressure measurement obtained with the s_1 mode of trampoline ii (see Figure 4).

Figure 5a shows the systematic errors, which are given by the relative deviations $\sigma_{S,Q} = Q_i/\tilde{Q}_{mod}(P_i) - 1$ and $\sigma_{S,f} = f_{m,i}/f_{COM}(P_i) - 1$. $\sigma_{S,Q}$ is within ±10%, except for one outlier at 3.9 mbar, which might originate from an increased measurement error of 35% of our commercial pressure gauge. Under this assumption (i.e., the actual pressure being 2.5 mbar instead of 3.9 mbar), the corresponding data points, both for Q and f_m (red dots in Figure 4b,c, respectively), would agree with the respective model functions $\tilde{Q}_{mod}$ and $f_{COM}/f_{COM,in} - 1$ (black dashed curve in Figure 4b and green dashed curve in Figure 4c, respectively). The systematic frequency error is shown in the pressure range of the VF regime, where f_m shows a pronounced P-dependency. Its absolute value reaches a minimum of about 2% at the FMF−VF transition; incomplete reflection of the trampoline's geometric imperfections and material properties in the Comsol simulation are considered to be limiting factors. Also, temperature or pressure variations in the sensor's vicinity might contribute. The independence of measured Q_{in} and f_{in} values from the incident optical power indicates no effects from laser-induced heating or dmaping.

Random errors $\sigma_{S,Q}$ and $\sigma_{R,f}$ are expressed as the standard deviation of repeated Q and f_m measurements, respectively, which are shown in Figure 5b. The values of $\sigma_{S,Q}$ obtained from ringdown measurements ($P < 10^{-2}$ mbar) are on average about 2-fold smaller than the values extracted from fits of the thermal noise spectrum ($P > 10^{-2}$ mbar). The random error of the f_m measurement, shown in the VF pressure range, is on average about 100-fold smaller than the one of the Q measurement.

While for the Q measurement, systematic and random errors contribute about equally, the f measurement is dominated by systematic errors.

To determine the overall error of the pressure values inferred from Q and f_m measurements, the following expressions are applied, respectively

$$\left(\frac{\Delta P}{P}\right)_Q = \left(\frac{\partial \tilde{Q}_{mod}}{\partial P}\right)^{-1} \frac{\tilde{Q}_{mod}}{P} \sqrt{\sigma_{S,Q}^2 + \sigma_{R,Q}^2}$$

$$\left(\frac{\Delta P}{P}\right)_f = \left(\frac{\partial f_{COM}}{\partial P}\right)^{-1} \frac{f_{COM}}{P} \sqrt{\sigma_{S,f}^2 + \sigma_{R,f}^2}$$

Here, the root sum square of systematic and random errors is scaled based on the corresponding model function and its derivative with respect to pressure. These scalings are shown in Figure 5c, where for the Q measurement, five distinct ranges can be identified. Between 10^{-7} and 10^{-5} mbar, where $\tilde{Q}_{mod}$ transitions from Q_{in} (constant) to $Q_{FMF} \propto 1/P$, the scaling reduces from 25 toward 1. Between 10^{-5} and 10^{-1} mbar, $\tilde{Q}_{mod} \approx Q_{FMF}$, which results in a scaling of 1. In the FMF−VF transition region, occurring approximately between 10^{-1} and 5 mbar, the scaling increases up to 3. It stays constant, up to about 10^2 mbar, where it starts further increasing toward ambient pressure. This increase is due to the flattening out of $\tilde{Q}_{mod}$ (see Figure 4b), which might be linked to Q approaching unity. The reasoning behind this assumption is that neither for the s_2 mode of trampoline ii (Figure 4b) nor for the s_1 or t_2 modes of trampoline i (see Figure 3a), this behavior is evident. For these modes, $Q > 10$ at ambient pressure. The error scaling related to f_m exceeds the one related to Q for most of the investigated pressure range. This reflects a comparatively reduced pressure sensitivity of f_m compared to that of Q.

Figure 5d shows the errors associated with pressure values inferred from Q and f_m measurements according to the expressions given above. The Q measurement results in lower errors for $P \leq 287$ mbar, whereas f_m yields lower errors for $P > 287$ mbar. Over a measurement range of 5-decades, 10^{-6} mbar $\lesssim P \lesssim 10^{-1}$ mbar, $\Delta P/P < 10\%$. At the low pressure end ($P \lesssim 10^{-6}$ mbar) and in the VF range ($P \gtrsim 10^{-1}$ mbar), the error increases, mainly due to reduced sensitivity. Overall, $\Delta P/P < 25\%$, except for the outliers at $P = \{3.9, 9.5\}$ mbar with $\Delta P/P = \{57, 29\}$ %.

Conclusions. In summary, we demonstrate a single nanomechanical gas pressure sensor, which exhibits, to the best of our knowledge, an unprecedented measurement range of 10-decades. Combining an analytical model for molecular damping with FEM simulations (Comsol) for viscous damping, enables matching the measured pressure dependency of our sensor's mechanical quality factor within 15%, over the investigated pressure range. This model thereby provides an accurate description of the transition between molecular and VF regimes. Furthermore, the FEM simulation reproduces the measured pressure dependency of the sensor's oscillation frequency in the VF regime to within 4%. Our model relies only on known membrane parameters and gas type, together with values of the sensor's intrinsic resonance frequency and quality factor, which are measured at a pressure $<10^{-7}$ mbar. This self-calibration of the sensor significantly reduces the effort required compared to calibrating it via point-by-point comparison with a reference gauge.[40] The error of the pressure values inferred from quality factor and frequency measurements lies within 10%, for pressures between $\sim 10^{-6}$ and $\sim 10^{-1}$ mbar, and within 25% for the complete measurement range from 10^{-7} mbar to ambient pressure. Exceptions are two data points at $P = \{3.9, 9.5\}$ mbar with related measurement errors of {57, 29} %. We point out that the established accuracy, particularly of these outliers, might be limited by our commercial pressure gauge, which suggests comparing our sensor to a more accurate reference gauge for determining its fundamental limitation. Also, we discuss the nonideal design of a membrane as a possible origin of deviations between data and model. A possible route toward increased pressure sensitivity, and, possibly, reduced measurement error, appears to be the exploitation of the squeeze-film effect, resulting from installing an oscillating membrane at a μm-scale distance from a neighboring surface.[51] Furthermore, the squeeze-film effect is expected to enable significantly (up to few orders of magnitude) increase the damping acting on the membrane.[52] This would translate to a decrease in the measurement time. Increasing our sensor's intrinsic quality factor, e.g., by following a similar approach to ref.,[23] enables further extending its measurement range to ultrahigh vacuum. Furthermore, we

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expect our sensor to be suitable for measuring pressures significantly higher than ambient pressure, based on corresponding investigations for compact cantilevers[53] (in our current setup, such investigations are prohibited by optical viewports incompatible with overpressure). This, together with its inherent compatibility with cryogenic environments,[54] temperatures of several 100 °C,[55] other gases than air, magnetic fields,[54] and spatially constrained environments (when combined, e.g., with a compact readout via an optical fiber,[56] on-chip photonics,[57] or electronics[58]), illustrates the sensor's versatility.

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C.R. and H.M. contributed equally to this work. C.R., H.M., J.S., A.L., and R.S. designed the research approach; H.M. built the experiment with the help of C.R. and S.C.; H.M. performed the measurements with the help of C.R.; C.R. developed the theoretical model; H.M., C.R., and N.S. analyzed the data; C.R., M.B.K.K., A.F., and H.M. contributed to the simulations; and C.R., A.L., and R.S. wrote the paper.

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ADDITIONAL NOTE

[a]Here and in the following beams, they are considered to be doubly clamped.

REFERENCES

(1) Nakhosteen, C. B.; Jousten, K. *Handbook of Vacuum Technology*; John Wiley & Sons, 2016.

(2) Shirhatti, V.; Nuthalapati, S.; Kedambaimoole, V.; Bhardwaj, A.; Nayak, M. M.; Rajanna, K. Broad-range fast response vacuum pressure sensors based on a graphene nanocomposite with hollow α-Fe2O3 microspheres. *ACS Appl. Electron. Mater.* **2020**, *2*, 2429−2439.

(3) Kim, T.; Ko, J.; Lee, J. Self-assembled silicon membrane resonator for high vacuum pressure sensing. *Vacuum* **2022**, *201*, 111101.

(4) Chen, Y.; Liu, S.; Hong, G.; Zou, M.; Liu, B.; Luo, J.; Wang, Y. Nano-optomechanical Resonators for Sensitive Pressure Sensing. *ACS Appl. Mater. Interfaces* **2022**, *14*, 39211−39219.

(5) Eaton, W. P.; Smith, J. H. Micromachined pressure sensors: review and recent developments. *Smart Mater. Struct.* **1997**, *6*, 530−539.

(6) Kokubun, K.; Hirata, M.; Murakami, H.; Toda, Y.; Ono, M. A bending and stretching mode crystal oscillator as a friction vacuum gauge. *Vacuum* **1984**, *34*, 731−735.

(7) Blom, F.; Bouwstra, S.; Elwenspoek, M.; Fluitman, J. Dependence of the quality factor of micromachined silicon beam resonators on pressure and geometry. *J. Vac. Sci. Technol., B: Microelectron. Nanometer Struct.–Process., Meas., Phenom.* **1992**, *10*, 19−26.

(8) Bianco, S.; Cocuzza, M.; Ferrero, S.; Giuri, E.; Piacenza, G.; Pirri, C.; Ricci, A.; Scaltrito, L.; Bich, D.; Merialdo, A.; et al. Silicon resonant microcantilevers for absolute pressure measurement. *J. Vac. Sci. Technol., B: Microelectron. Nanometer Struct.–Process., Meas., Phenom.* **2006**, *24*, 1803−1809.

(9) Li, M.; Tang, H. X.; Roukes, M. L. Ultra-sensitive NEMS-based cantilevers for sensing, scanned probe and very high-frequency applications. *Nat. Nanotechnol.* **2007**, *2*, 114−120.

(10) Martin, M. J.; Houston, B. H. Gas damping of carbon nanotube oscillators. *Appl. Phys. Lett.* **2007**, *91*, 103116.

(11) Southworth, D.; Craighead, H. G.; Parpia, J. Pressure dependent resonant frequency of micromechanical drumhead resonators. *Appl. Phys. Lett.* **2009**, *94*, 213506.

(12) Lübbe, J.; Temmen, M.; Schnieder, H.; Reichling, M. Measurement and modelling of non-contact atomic force microscope cantilever properties from ultra-high vacuum to normal pressure conditions. *Meas. Sci. Technol.* **2011**, *22*, 055501.

(13) Smith, A.; Niklaus, F.; Paussa, A.; Vaziri, S.; Fischer, A. C.; Sterner, M.; Forsberg, F.; Delin, A.; Esseni, D.; Palestri, P.; et al. Electromechanical Piezoresistive Sensing in Suspended Graphene Membranes. *Nano Lett.* **2013**, *13*, 3237−3242.

(14) Lee, J.; Wang, Z.; He, K.; Shan, J.; Feng, P. X.-L. Air damping of atomically thin MoS2 nanomechanical resonators. *Appl. Phys. Lett.* **2014**, *105*, 023104.

(15) Dolleman, R. J.; Davidovikj, D.; Cartamil-Bueno, S. J.; van der Zant, H. S.; Steeneken, P. G. Graphene squeeze-film pressure sensors. *Nano Lett.* **2016**, *16*, 568−571.

(16) Smith, A. D.; Niklaus, F.; Paussa, A.; Schröder, S.; Fischer, A. C.; Sterner, M.; Wagner, S.; Vaziri, S.; Forsberg, F.; Esseni, D.; et al. Piezoresistive Properties of Suspended Graphene Membranes under Uniaxial and Biaxial Strain in Nanoelectromechanical Pressure Sensors. *ACS Nano* **2016**, *10*, 9879−9886.

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(17) Naesby, A.; Naserbakht, S.; Dantan, A. Effects of pressure on suspended micromechanical membrane arrays. *Appl. Phys. Lett.* **2017**, *111*, 201103.
(18) Wagner, S.; Yim, C.; McEvoy, N.; Kataria, S.; Yokaribas, V.; Kuc, A.; Pindl, S.; Fritzen, C.-P.; Heine, T.; Duesberg, G. S.; et al. Highly Sensitive Electromechanical Piezoresistive Pressure Sensors Based on Large-Area Layered $PtSe_2$ Films. *Nano Lett.* **2018**, *18*, 3738−3745.
(19) Alcheikh, N.; Hajjaj, A.; Younis, M. I. Highly sensitive and wide-range resonant pressure sensor based on the veering phenomenon. *Sens. Actuators, A* **2019**, *300*, 111652.
(20) Song, P.; Ma, Z.; Ma, J.; Yang, L.; Wei, J.; Zhao, Y.; Zhang, M.; Yang, F.; Wang, X. Recent progress of miniature MEMS pressure sensors. *Micromachines* **2020**, *11*, 56.
(21) Ghatge, M.; Walters, G.; Nishida, T.; Tabrizian, R. A 30-nm thick integrated hafnium zirconium oxide nano-electro-mechanical membrane resonator. *Appl. Phys. Lett.* **2020**, *116*, 043501.
(22) Reinhardt, C. *Ultralow-noise Silicon Nitride Trampoline Resonators for Sensing and Optomechanics*; McGill University: Canada, 2018.
(23) Saarinen, S. A.; Kralj, N.; Langman, E. C.; Tsaturyan, Y.; Schliesser, A. Laser cooling a membrane-in-the-middle system close to the quantum ground state from room temperature. *Optica* **2023**, *10*, 364−372.
(24) Christian, R. The theory of oscillating-vane vacuum gauges. *Vacuum* **1966**, *16*, 175−178.
(25) Newell, W. E. Miniaturization of Tuning Forks: Integrated electronic circuits provide the incentive and the means for orders-of-magnitude reduction in size. *Science* **1968**, *161*, 1320−1326.
(26) Langdon, R. M. Resonator sensors-a review. *J. Phys. E: Sci. Instrum.* **1985**, *18*, 103−115.
(27) Sader, J. E. Frequency response of cantilever beams immersed in viscous fluids with applications to the atomic force microscope. *J. Appl. Phys.* **1998**, *84*, 64−76.
(28) Yadykin, Y.; Tenetov, V.; Levin, D. The added mass of a flexible plate oscillating in a fluid. *J. Fluid Struct.* **2003**, *17*, 115−123.
(29) Verbridge, S. S.; Ilic, R.; Craighead, H. G.; Parpia, J. M. Size and frequency dependent gas damping of nanomechanical resonators. *Appl. Phys. Lett.* **2008**, *93*, 013101.
(30) Kara, V.; Sohn, Y.-I.; Atikian, H.; Yakhot, V.; Loncar, M.; Ekinci, K. L. Nanofluidics of single-crystal diamond nanomechanical resonators. *Nano Lett.* **2015**, *15*, 8070−8076.
(31) Bhiladvala, R. B.; Wang, Z. J. Effect of fluids on the Q factor and resonance frequency of oscillating micrometer and nanometer scale beams. *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.* **2004**, *69*, 036307.
(32) Ghatkesar, M. K.; Braun, T.; Barwich, V.; Ramseyer, J.-P.; Gerber, C.; Hegner, M.; Lang, H. P. Resonating modes of vibrating microcantilevers in liquid. *Appl. Phys. Lett.* **2008**, *92*, 043106.
(33) Hosaka, H.; Itao, K.; Kuroda, S. Damping characteristics of beam-shaped micro-oscillators. *Sens. Actuators, A* **1995**, *49*, 87−95.
(34) Landau, L. D.; Lifshitz, E. M. *Fluid Mechanics: Landau and Lifshitz: Course of Theoretical Physics*; Elsevier, 2013; Vol. *6*.
(35) Karabacak, D.; Yakhot, V.; Ekinci, K. High-frequency nanofluidics: An experimental study using nanomechanical resonators. *Phys. Rev. Lett.* **2007**, *98*, 254505.
(36) Yakhot, V.; Colosqui, C. Stokes' second flow problem in a high-frequency limit: application to nanomechanical resonators. *J. Fluid Mech.* **2007**, *586*, 249−258.
(37) Liem, A. T.; Ari, A. B.; Ti, C.; Cops, M. J.; McDaniel, J. G.; Ekinci, K. L. Nanoflows induced by MEMS and NEMS: Limits of two-dimensional models. *Phys. Rev. Fluids* **2021**, *6*, 024201.
(38) Reinhardt, C.; Müller, T.; Bourassa, A.; Sankey, J. C. Ultralow-noise SiN trampoline resonators for sensing and optomechanics. *Phys. Rev. X* **2016**, *6*, 021001.
(39) Norte, R. A.; Moura, J. P.; Gröblacher, S. Mechanical resonators for quantum optomechanics experiments at room temperature. *Phys. Rev. Lett.* **2016**, *116*, 147202.
(40) ISO 3567 (2011) Vacuum Gauges—Calibration by direct comparison with a reference gauge.
(41) Jousten, K.; Boineau, F.; Bundaleski, N.; Illgen, C.; Setina, J.; Teodoro, O. M.; Vicar, M.; Wüest, M. A review on hot cathode ionisation gauges with focus on a suitable design for measurement accuracy and stability. *Vacuum* **2020**, *179*, 109545.
(42) Scherschligt, J.; Fedchak, J. A.; Barker, D. S.; Eckel, S.; Klimov, N.; Makrides, C.; Tiesinga, E. Development of a new UHV/XHV pressure standard (cold atom vacuum standard). *Metrologia* **2017**, *54*, S125−S132.
(43) Norcada, A. MEMS Technology Company. https://www.norcada.com/ (accessed Feb 13, 2024).
(44) Wilson, D. J.; Regal, C. A.; Papp, S. B.; Kimble, H. Cavity optomechanics with stoichiometric SiN films. *Phys. Rev. Lett.* **2009**, *103*, 207204.
(45) Yu, P.-L.; Purdy, T.; Regal, C. Control of material damping in high-Q membrane microresonators. *Phys. Rev. Lett.* **2012**, *108*, 083603.
(46) Tsaturyan, Y.; Barg, A.; Polzik, E. S.; Schliesser, A. Ultracoherent nanomechanical resonators via soft clamping and dissipation dilution. *Nat. Nanotechnol.* **2017**, *12*, 776−783.
(47) Villanueva, L. G.; Schmid, S. Evidence of surface loss as ubiquitous limiting damping mechanism in SiN micro-and nano-mechanical resonators. *Phys. Rev. Lett.* **2014**, *113*, 227201.
(48) Høj, D.; Wang, F.; Gao, W.; Hoff, U. B.; Sigmund, O.; Andersen, U. L. Ultra-coherent nanomechanical resonators based on inverse design. *Nat. Commun.* **2021**, *12*, 5766.
(49) Pfeiffer PBR 260—FullRange Pirani/Bayard-Alpert-Transmitter. https://www.pfeiffer-vacuum.com/de/download-center/download!download.action?request_locale=de_DE&referer=2063&download=/filepool/file/measurement/bg5131ben_b.pdf, (accessed Feb 13, 2024).
(50) Bereyhi, M. J.; Beccari, A.; Groth, R.; Fedorov, S. A.; Arabmoheghi, A.; Kippenberg, T. J.; Engelsen, N. J. Hierarchical tensile structures with ultralow mechanical dissipation. *Nat. Commun.* **2022**, *13*, 3097.
(51) Salimi, M.; Nielsen, R. V.; Pedersen, H. B.; Dantan, A. Squeeze film absolute pressure sensors with sub-millipascal sensitivity. *arXiv* **2023**, arXiv:2312.11915. (accessed March 5, 2024)
(52) Suijlen, M.; Koning, J.; van Gils, M.; Beijerinck, H. Squeeze film damping in the free molecular flow regime with full thermal accommodation. *Sens. Actuators, A* **2009**, *156*, 171−179.
(53) Svitelskiy, O.; Sauer, V.; Liu, N.; Cheng, K.-M.; Finley, E.; Freeman, M. R.; Hiebert, W. K. Pressurized fluid damping of nanoelectromechanical systems. *Phys. Rev. Lett.* **2009**, *103*, 244501.
(54) Zwickl, B. M.; Shanks, W. E.; Jayich, A. M.; Yang, C.; Bleszynski Jayich, A. C.; Thompson, J. D.; Harris, J. G. E. High quality mechanical and optical properties of commercial silicon nitride membranes. *Appl. Phys. Lett.* **2008**, *92*, 103125.
(55) St-Gelais, R.; Bernard, S.; Reinhardt, C.; Sankey, J. C. Swept-frequency drumhead optomechanical resonators. *ACS Photonics* **2019**, *6*, 525−530.
(56) Flowers-Jacobs, N. E.; Hoch, S. W.; Sankey, J. C.; Kashkanova, A.; Jayich, A. M.; Deutsch, C.; Reichel, J.; Harris, J. G. E. Fiber-cavity-based optomechanical device. *Appl. Phys. Lett.* **2012**, *101*, 221109.
(57) Guo, J.; Gröblacher, S. Integrated optical-readout of a high-Q mechanical out-of-plane mode. *Light: Sci. Appl.* **2022**, *11*, 282.
(58) Bagci, T.; Simonsen, A.; Schmid, S.; Villanueva, L. G.; Zeuthen, E.; Appel, J.; Taylor, J. M.; Sørensen, A.; Usami, K.; Schliesser, A.; et al. Optical detection of radio waves through a nanomechanical transducer. *Nature* **2014**, *507*, 81−85.

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